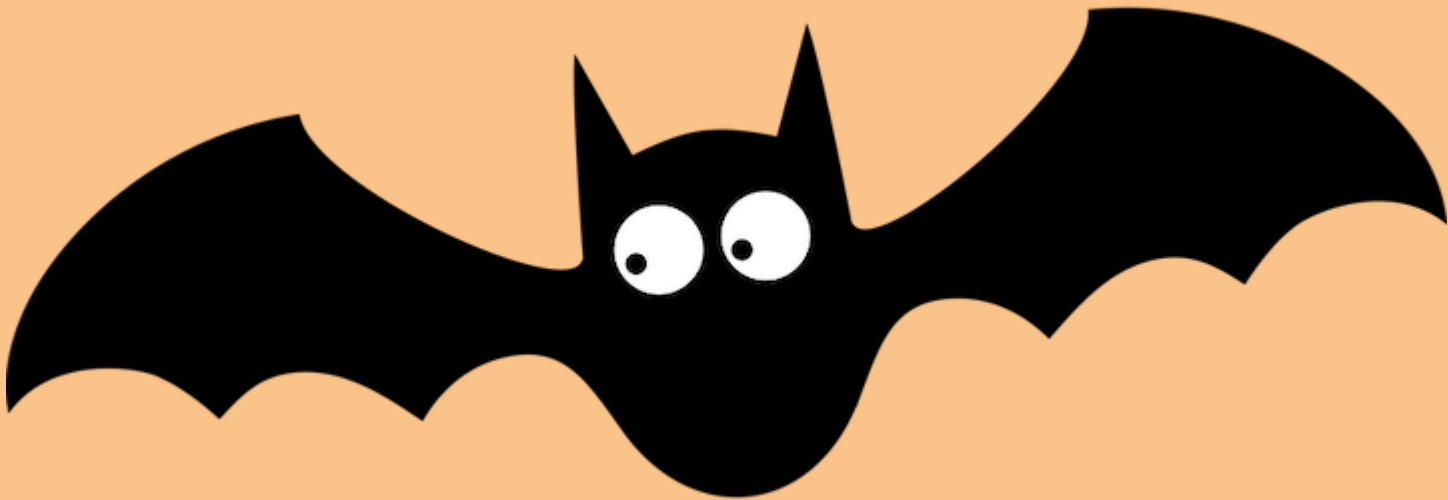




Adaboost

Son Nguyen

Adaboost

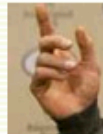
Idea Behind Ada Boost

- Examples of high weight are shown more often at later rounds
- Face/nonface classification problem:

Round 1

best weak classifier:

change weights:

						
$1/7$	$1/7$	$1/7$	$1/7$	$1/7$	$1/7$	$1/7$
✓	✗	✓	✓	✗	✓	✗
$1/16$	$1/4$	$1/16$	$1/16$	$1/4$	$1/16$	$1/4$

Round 2

best weak classifier:

change weights:

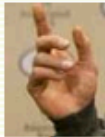
									
✓	✓	✓	✗	✗	✗	✓	✓	✓	✓
$1/8$	$1/32$	$1/32$	$11/32$		$1/2$		$1/8$	$1/32$	$1/32$

Adaboost

Idea Behind Ada Boost

Round 3



- out of all available weak classifiers, we choose the one that works best on the data we have at round 3
- we assume there is always a weak classifier better than random (better than 50% error)
-  image is half of the data given to the classifier
- chosen weak classifier **has to** classify this image correctly

Adaboost, Clearly Explained

- Demonstration by StatQuest
- [Link](#)

Calculation Example

Data

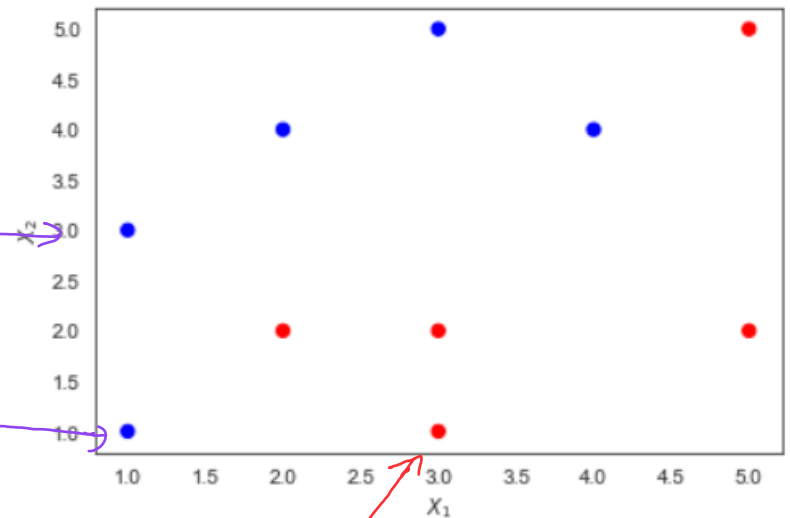
x_1	x_2	y
1	1	1
1	3	1
2	2	-1
2	4	1
3	1	-1
3	2	-1
3	5	1
4	4	1
5	2	-1
5	5	-1

Calculation Example

Data

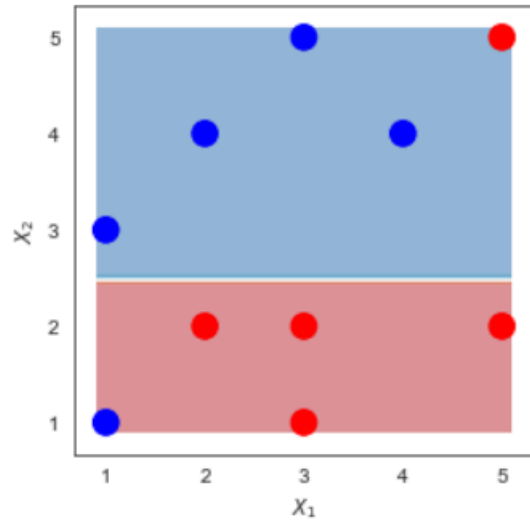
Blue = 1 ; Red = -1

x_1	x_2	y
1	1	1
1	3	1
2	2	-1
2	4	1
3	1	-1
3	2	-1
3	5	1
4	4	1
5	2	-1
5	5	-1

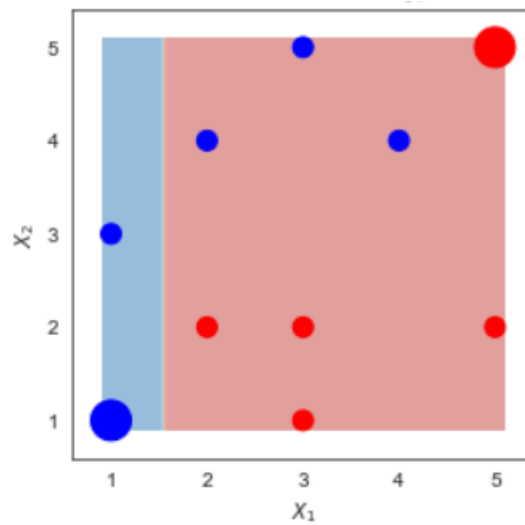
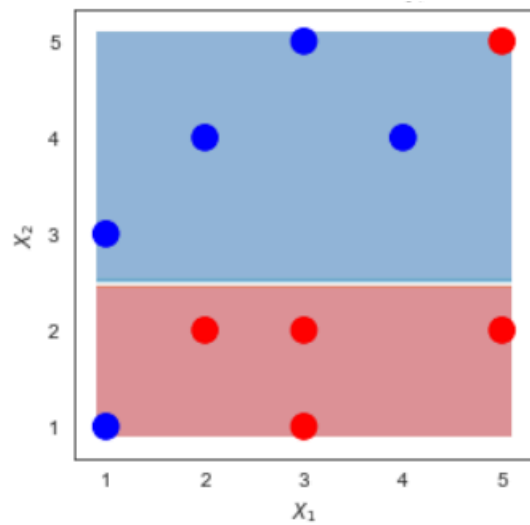


Adaboost in a nutshell

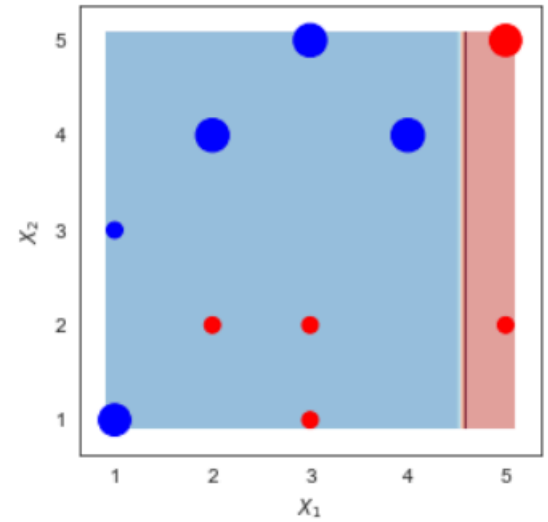
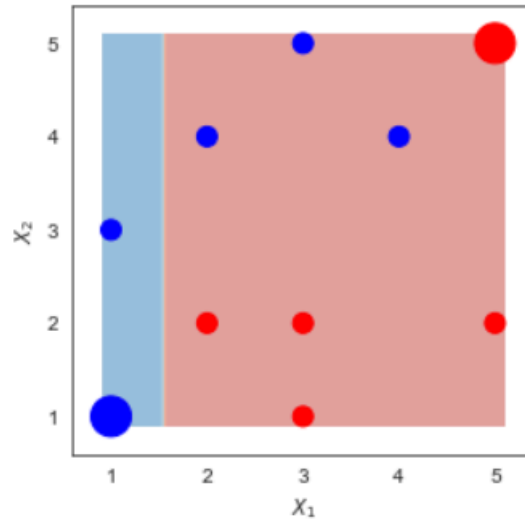
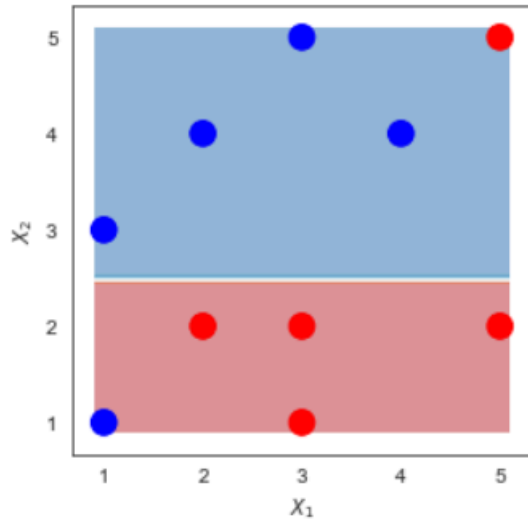
Make Stump 1



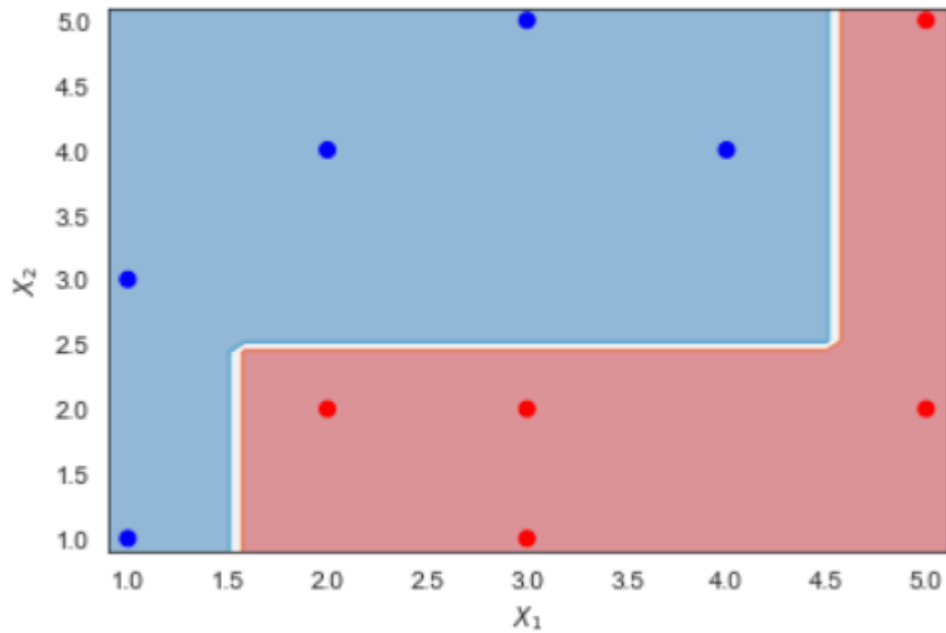
Make Stump 2



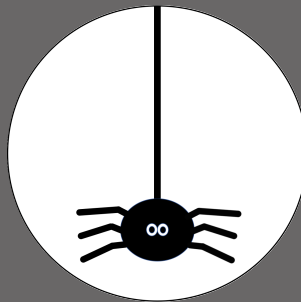
Make Stump 3



Combine the Stumps



Detail Calculation



Make the first stump

Row	x1	x2	y
0	1	1	1
1	1	3	1
2	2	2	-1
3	2	4	1
4	3	1	-1
5	3	2	-1
6	3	5	1
7	4	4	1
8	5	2	-1
9	5	5	-1

Make the first stump

- Assign weights for each row
- Every row has the same weight in the first step

Row	x1	x2	y
0	1	1	1
1	1	3	1
2	2	2	-1
3	2	4	1
4	3	1	-1
5	3	2	-1
6	3	5	1
7	4	4	1
8	5	2	-1
9	5	5	-1

Make the first stump

- Assign weights for each row
- Every row has the same weight in the first step

Row	x1	x2	y	Weight 1
0	1	1	1	0.1
1	1	3	1	0.1
2	2	2	-1	0.1
3	2	4	1	0.1
4	3	1	-1	0.1
5	3	2	-1	0.1
6	3	5	1	0.1
7	4	4	1	0.1
8	5	2	-1	0.1
9	5	5	-1	0.1

Make the first stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits

Row	x1	x2	y	Weight 1
0	1	1	1	0.1
1	1	3	1	0.1
2	2	2	-1	0.1
3	2	4	1	0.1
4	3	1	-1	0.1
5	3	2	-1	0.1
6	3	5	1	0.1
7	4	4	1	0.1
8	5	2	-1	0.1
9	5	5	-1	0.1

Make the first stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits
- The split with the lowest children impurity is the best split

Row	x1	x2	y	Weight 1
0	1	1	1	0.1
1	1	3	1	0.1
2	2	2	-1	0.1
3	2	4	1	0.1
4	3	1	-1	0.1
5	3	2	-1	0.1
6	3	5	1	0.1
7	4	4	1	0.1
8	5	2	-1	0.1
9	5	5	-1	0.1

Make the first stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits
- The split with the lowest children impurity is the best split
- **NOTE:** You are not required to make the stump. So the stump will be given to you!

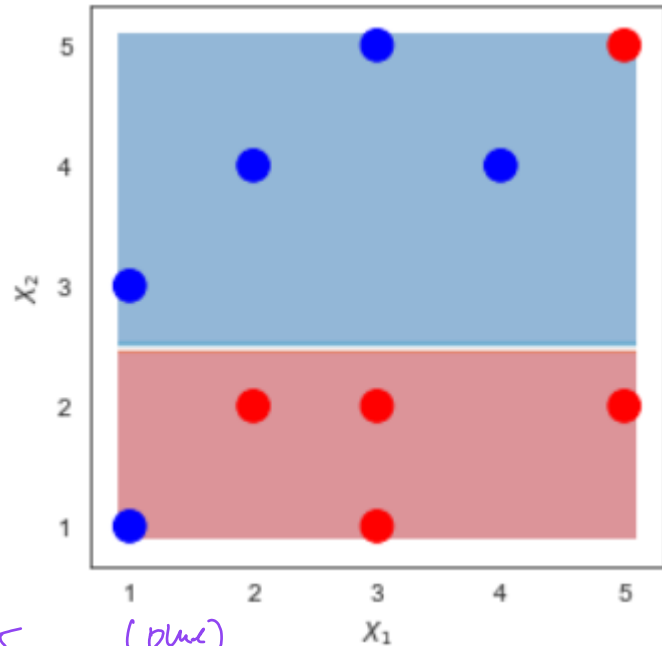


Row	x1	x2	y	Weight 1
0	1	1	1	0.1
1	1	3	1	0.1
2	2	2	-1	0.1
3	2	4	1	0.1
4	3	1	-1	0.1
5	3	2	-1	0.1
6	3	5	1	0.1
7	4	4	1	0.1
8	5	2	-1	0.1
9	5	5	-1	0.1

Make the first stump

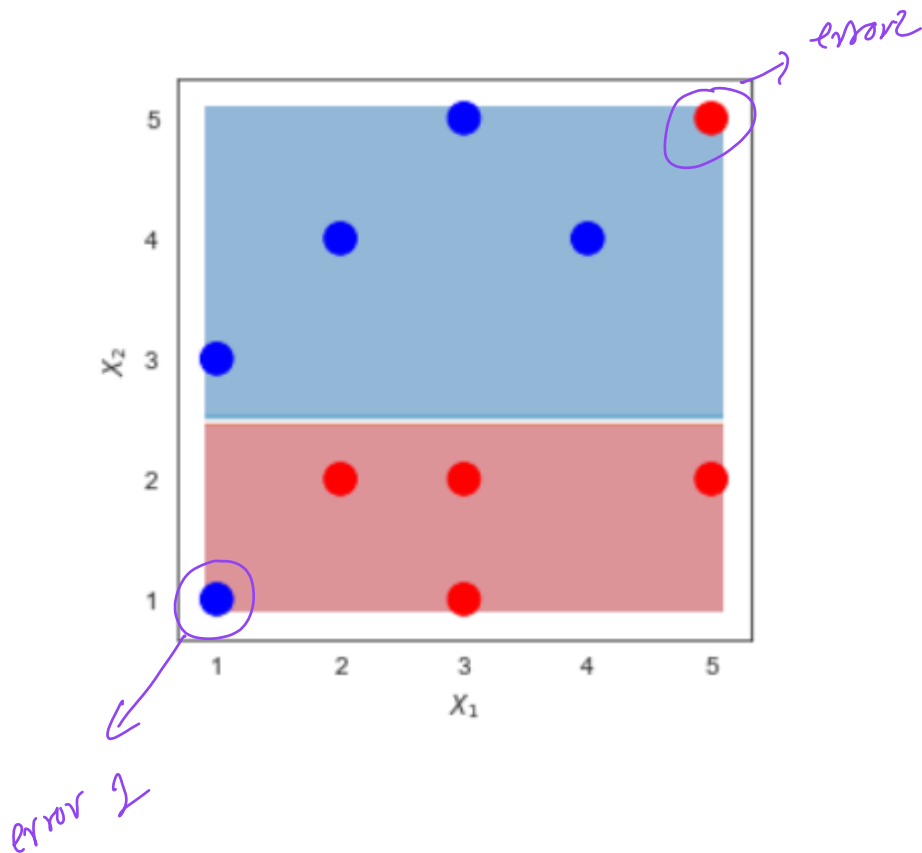
- Use Weighted Gini-Index to calculate the children entropy of all candidate splits
- The split with the lowest children impurity is the best split
- **NOTE:** You are not required to make the stump. So the stump will be given to you!
- Here is the first stump

$$I(x_2 > 2.5) = \begin{cases} 1 & \text{if } x_2 > 2.5 \quad (\text{blue}) \\ -1 & \text{if } x_2 < 2.5 \quad (\text{red}) \end{cases}$$



Make the first stump

- **Stump 1:** $I(x_2 > 2.5)$



Prediction of Stump 1

- **Stump 1:**

$$I(x_2 > 2.5)$$

- If $x_2 > 2.5$, predicts $y = 1$.
- Otherwise, predicts $y = -1$

Row	x1	x2	y	Stump 1 Predicts
0	1	1	1	-1 <i>error 1</i>
1	1	3	1	1
2	2	2	-1	-1
3	2	4	1	1
4	3	1	-1	-1
5	3	2	-1	-1
6	3	5	1	1
7	4	4	1	1
8	5	2	-1	-1
9	5	5	-1	1 <i>error 2</i>

Error of the first stump

Row	x1	x2	y	Stump 1 Predicts	Weight 1	
0	1	1	1	-1	0.1	<-
1	1	3	1	1	0.1	
2	2	2	-1	-1	0.1	
3	2	4	1	1	0.1	
4	3	1	-1	-1	0.1	
5	3	2	-1	-1	0.1	
6	3	5	1	1	0.1	
7	4	4	1	1	0.1	
8	5	2	-1	-1	0.1	
9	5	5	-1	1	0.1	<-

Error of the first stump

- Stump 1 has 2 misclassifications at row 0 and 9 (The predictions are NOT the same as the y values). The total weights of these rows are:

$$\epsilon_1 = 0.1 + 0.1 = 0.2$$

Row	x1	x2	y	Stump 1 Predicts	Weight 1	
0	1	1	1	-1	0.1	<-
1	1	3	1	1	0.1	
2	2	2	-1	-1	0.1	
3	2	4	1	1	0.1	
4	3	1	-1	-1	0.1	
5	3	2	-1	-1	0.1	
6	3	5	1	1	0.1	
7	4	4	1	1	0.1	
8	5	2	-1	-1	0.1	
9	5	5	-1	1	0.1	<-

Voting Power of the first Stump

- Stump 1 has 2 misclassifications at row 0 and 9 (The predictions are NOT the same as the y values). The total weights of these rows are:

$$\epsilon_1 = 0.1 + 0.1 = 0.2$$

- Voting Power: (L is the learning rate. $L = 1$ in this example 1)

$$\alpha_1 = L \cdot \frac{1}{2} \cdot \ln\left(\frac{1 - \epsilon_1}{\epsilon_1}\right) = 0.693$$

$L \rightarrow \Rightarrow \propto \rightarrow \Rightarrow$

Row	x1	x2	y	Stump 1 Predicts	Weight 1	
0	1	1	1	-1	0.1	<-
1	1	3	1	1	0.1	
2	2	2	-1	-1	0.1	
3	2	4	1	1	0.1	
4	3	1	-1	-1	0.1	
5	3	2	-1	-1	0.1	
6	3	5	1	1	0.1	
7	4	4	1	1	0.1	
8	5	2	-1	-1	0.1	
9	5	5	-1	1	0.1	<-

Calculating the new weights

- For misclassified rows 0 and 9:

$$w_{new} = w_{old} \cdot e^{\alpha} = 0.1 \cdot e^{0.693} = 0.2$$

- For the correctly classified rows:

$$w_{new} = w_{old} \cdot e^{-\alpha} = 0.1 \cdot e^{-0.693} = .05$$

$$L \downarrow \Rightarrow \alpha \downarrow \Rightarrow e^{\alpha} \downarrow$$

$\rightarrow w_{new}$ of misclassified obs. $\downarrow$

$e^{-\alpha} \nearrow \Rightarrow w_{new}$ of correctly classified $\nearrow$

Row	x1	x2	y	Stump 1 Predicts	Weight 1	
0	1	1	1	-1	0.1	<-
1	1	3	1	1	0.1	
2	2	2	-1	-1	0.1	
3	2	4	1	1	0.1	
4	3	1	-1	-1	0.1	
5	3	2	-1	-1	0.1	
6	3	5	1	1	0.1	
7	4	4	1	1	0.1	
8	5	2	-1	-1	0.1	
9	5	5	-1	1	0.1	<-

Calculating the new weights

- For misclassified rows 0 and 9:

$$w_{new} = w_{old} \cdot e^{\alpha} = 0.1 \cdot e^{0.693} = 0.2$$

- For the correctly classified rows:

$$w_{new} = w_{old} \cdot e^{-\alpha} = 0.1 \cdot e^{-0.693} = .05$$

Row	x1	x2	y	Stump 1 Predicts	Weight 1	Weight 2
0	1	1	1	-1	0.1	0.2
1	1	3	1	1	0.1	0.05
2	2	2	-1	-1	0.1	0.05
3	2	4	1	1	0.1	0.05
4	3	1	-1	-1	0.1	0.05
5	3	2	-1	-1	0.1	0.05
6	3	5	1	1	0.1	0.05
7	4	4	1	1	0.1	0.05
8	5	2	-1	-1	0.1	0.05
9	5	5	-1	1	0.1	0.2

Calculating the new weights

- For misclassified rows 0 and 9:

$$w_{new} = w_{old} \cdot e^{\alpha} = 0.1 \cdot e^{0.693} = 0.2$$

- For the correctly classified rows:

$$w_{new} = w_{old} \cdot e^{-\alpha} = 0.1 \cdot e^{-0.693} = .05$$

- Notice how the weights increase for misclassified rows and decrease otherwise.

Row	x1	x2	y	Stump 1 Predicts	Weight 1	Weight 2
0	1	1	1	-1	0.1	0.2
1	1	3	1	1	0.1	0.05
2	2	2	-1	-1	0.1	0.05
3	2	4	1	1	0.1	0.05
4	3	1	-1	-1	0.1	0.05
5	3	2	-1	-1	0.1	0.05
6	3	5	1	1	0.1	0.05
7	4	4	1	1	0.1	0.05
8	5	2	-1	-1	0.1	0.05
9	5	5	-1	1	0.1	0.2

Calculating the new weights

- The total weights has to be 1. We divide the weights by the total ($.2 * 2 + .05 * 8 = .8$) to achieve this.

Row	x1	x2	y	Stump 1 Predicts	Weight 1	Weight 2
0	1	1	1	-1	0.1	0.2
1	1	3	1	1	0.1	0.05
2	2	2	-1	-1	0.1	0.05
3	2	4	1	1	0.1	0.05
4	3	1	-1	-1	0.1	0.05
5	3	2	-1	-1	0.1	0.05
6	3	5	1	1	0.1	0.05
7	4	4	1	1	0.1	0.05
8	5	2	-1	-1	0.1	0.05
9	5	5	-1	1	0.1	0.2

Calculating the new weights

- The total weights has to be 1. We divide the weights by the total ($.2 * 2 + .05 * 8 = .8$) to achieve this.
- Divide Weight 2 by 0.8

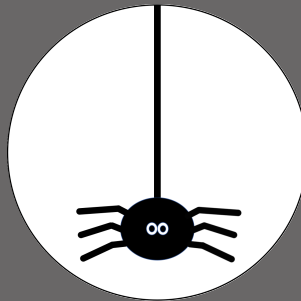
Row	x1	x2	y	Stump 1 Predicts	Weight 1	Weight 2
0	1	1	1	-1	0.1	0.2
1	1	3	1	1	0.1	0.05
2	2	2	-1	-1	0.1	0.05
3	2	4	1	1	0.1	0.05
4	3	1	-1	-1	0.1	0.05
5	3	2	-1	-1	0.1	0.05
6	3	5	1	1	0.1	0.05
7	4	4	1	1	0.1	0.05
8	5	2	-1	-1	0.1	0.05
9	5	5	-1	1	0.1	0.2

Calculating the new weights

- The total weights has to be 1. We divide the weights by the total ($.2 * 2 + .05 * 8 = .8$) to achieve this.
- Divide Weight 2 by 0.8

Row	x1	x2	y	Stump 1 Predicts	Weight 1	Weight 2
0	1	1	1	-1	0.1	0.25
1	1	3	1	1	0.1	0.0625
2	2	2	-1	-1	0.1	0.0625
3	2	4	1	1	0.1	0.0625
4	3	1	-1	-1	0.1	0.0625
5	3	2	-1	-1	0.1	0.0625
6	3	5	1	1	0.1	0.0625
7	4	4	1	1	0.1	0.0625
8	5	2	-1	-1	0.1	0.0625
9	5	5	-1	1	0.1	0.25

Repeat the process to make the second
Stump



Data to make the second Stump

Row	x1	x2	y	Weight 2
0	1	1	1	0.25
1	1	3	1	0.0625
2	2	2	-1	0.0625
3	2	4	1	0.0625
4	3	1	-1	0.0625
5	3	2	-1	0.0625
6	3	5	1	0.0625
7	4	4	1	0.0625
8	5	2	-1	0.0625
9	5	5	-1	0.25

Make the second stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits

Row	x1	x2	y	Weight 2
0	1	1	1	0.25
1	1	3	1	0.0625
2	2	2	-1	0.0625
3	2	4	1	0.0625
4	3	1	-1	0.0625
5	3	2	-1	0.0625
6	3	5	1	0.0625
7	4	4	1	0.0625
8	5	2	-1	0.0625
9	5	5	-1	0.25

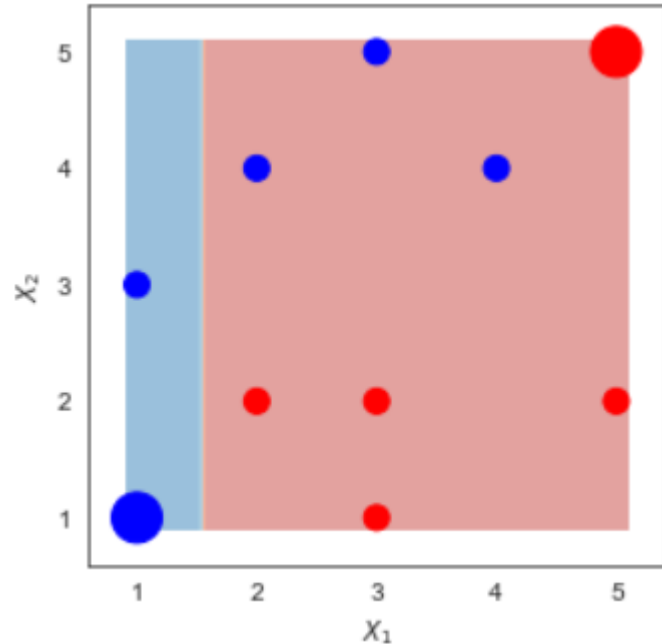
Make the second stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits
- The split with the lowest children impurity is the best split

Row	x1	x2	y	Weight 2
0	1	1	1	0.25
1	1	3	1	0.0625
2	2	2	-1	0.0625
3	2	4	1	0.0625
4	3	1	-1	0.0625
5	3	2	-1	0.0625
6	3	5	1	0.0625
7	4	4	1	0.0625
8	5	2	-1	0.0625
9	5	5	-1	0.25

Make the second stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits
- The split with the lowest children impurity is the best split



Error of the second stump

Row	x1	x2	y	Stump 2 Predicts	Weight 2	
0	1	1	1	1	0.25	
1	1	3	1	1	0.0625	
2	2	2	-1	-1	0.0625	
3	2	4	1	-1	0.0625	<-
4	3	1	-1	-1	0.0625	
5	3	2	-1	-1	0.0625	
6	3	5	1	-1	0.0625	<-
7	4	4	1	-1	0.0625	<-
8	5	2	-1	-1	0.0625	
9	5	5	-1	-1	0.25	

Error of the second stump

- Stump 2 has misclassifications at row 3, 6, and 7 (The predictions are NOT the same as the y values). The total weights of these rows are: $0.0625 + 0.0625 + 0.0625 = 0.1875$

- Error of Stump 2:

$$\epsilon_2 = 0.1875$$

- Voting Power:

$$\alpha_2 = L \cdot \frac{1}{2} \cdot \ln\left(\frac{1 - \epsilon_2}{\epsilon_2}\right) = 0.733$$

Row	x1	x2	y	Stump 2 Predicts	Weight 2	
0	1	1	1	1	0.25	
1	1	3	1	1	0.0625	
2	2	2	-1	-1	0.0625	
3	2	4	1	-1	0.0625	<-
4	3	1	-1	-1	0.0625	
5	3	2	-1	-1	0.0625	
6	3	5	1	-1	0.0625	<-
7	4	4	1	-1	0.0625	<-
8	5	2	-1	-1	0.0625	
9	5	5	-1	-1	0.25	

Calculating the new weights

- For misclassified rows 3, 6 and 7:

$$w_{new} = w_{old} \cdot e^{\alpha}$$

- For the correctly classified rows:

$$w_{new} = w_{old} \cdot e^{-\alpha}$$

Row	x1	x2	y	Stump 2 Predicts	Weight 2	
0	1	1	1	1	0.25	
1	1	3	1	1	0.0625	
2	2	2	-1	-1	0.0625	
3	2	4	1	-1	0.0625	<-
4	3	1	-1	-1	0.0625	
5	3	2	-1	-1	0.0625	
6	3	5	1	-1	0.0625	<-
7	4	4	1	-1	0.0625	<-
8	5	2	-1	-1	0.0625	
9	5	5	-1	-1	0.25	

Calculating the new weights

- For misclassified rows 3, 6 and 7:

$$w_{new} = w_{old} \cdot e^{\alpha}$$

- For the correctly classified rows:

$$w_{new} = w_{old} \cdot e^{-\alpha}$$

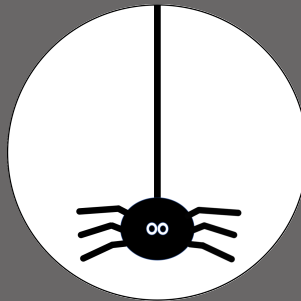
Row	x1	x2	y	Weight 2	Stump 2 Predicts	Weight 3
0	1	1	1	0.25	1	0.12012
1	1	3	1	0.0625	1	0.03003
2	2	2	-1	0.0625	-1	0.03003
3	2	4	1	0.0625	-1	0.13008
4	3	1	-1	0.0625	-1	0.03003
5	3	2	-1	0.0625	-1	0.03003
6	3	5	1	0.0625	-1	0.13008
7	4	4	1	0.0625	-1	0.13008
8	5	2	-1	0.0625	-1	0.03003
9	5	5	-1	0.25	-1	0.12012

Normalize the new weights

- The total weights has to be 1. We divide Weight 3 by the total of current Weight 3, which is 0.780624761 to achieve this.

Row	x1	x2	y	Weight 2	Stump 2 Predicts	Weight 3
0	1	1	1	0.25	1	0.15387
1	1	3	1	0.0625	1	0.03847
2	2	2	-1	0.0625	-1	0.03847
3	2	4	1	0.0625	-1	0.16664
4	3	1	-1	0.0625	-1	0.03847
5	3	2	-1	0.0625	-1	0.03847
6	3	5	1	0.0625	-1	0.16664
7	4	4	1	0.0625	-1	0.16664
8	5	2	-1	0.0625	-1	0.03847
9	5	5	-1	0.25	-1	0.15387

Repeat the process to make the third
Stump



Data to Make the third stump

Row	x1	x2	y	Weight 3
0	1	1	1	0.15387
1	1	3	1	0.03847
2	2	2	-1	0.03847
3	2	4	1	0.16664
4	3	1	-1	0.03847
5	3	2	-1	0.03847
6	3	5	1	0.16664
7	4	4	1	0.16664
8	5	2	-1	0.03847
9	5	5	-1	0.15387

Make the third stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits

Row	x1	x2	y	Weight 3
0	1	1	1	0.15387
1	1	3	1	0.03847
2	2	2	-1	0.03847
3	2	4	1	0.16664
4	3	1	-1	0.03847
5	3	2	-1	0.03847
6	3	5	1	0.16664
7	4	4	1	0.16664
8	5	2	-1	0.03847
9	5	5	-1	0.15387

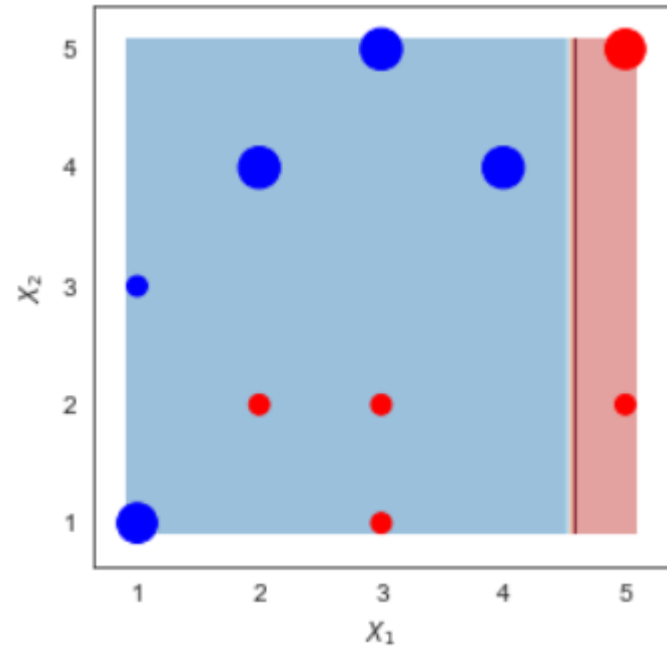
Make the third stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits
- The split with the lowest children impurity is the best split

Row	x1	x2	y	Weight 3
0	1	1	1	0.15387
1	1	3	1	0.03847
2	2	2	-1	0.03847
3	2	4	1	0.16664
4	3	1	-1	0.03847
5	3	2	-1	0.03847
6	3	5	1	0.16664
7	4	4	1	0.16664
8	5	2	-1	0.03847
9	5	5	-1	0.15387

Make the third stump

- Use Weighted Gini-Index to calculate the children entropy of all candidate splits
- The split with the lowest children impurity is the best split



Error of the third stump

Row	x1	x2	y	Stump 3 Predicts	Weight 3	
0	1	1	1	1	0.15385	
1	1	3	1	1	0.03846	
2	2	2	-1	1	0.03846	<-
3	2	4	1	1	0.16667	
4	3	1	-1	1	0.03846	<-
5	3	2	-1	1	0.03846	<-
6	3	5	1	1	0.16667	
7	4	4	1	1	0.16667	
8	5	2	-1	-1	0.03846	
9	5	5	-1	-1	0.15385	

Error of the third stump

- Stump 3 has misclassifications at row 2, 4, and 5 (The predictions are NOT the same as the y values). The total weights of these rows are:

$$\epsilon_3 = 0.03846 \cdot 3 = 0.11538$$

- Voting Power:

$$\alpha_3 = L \cdot \frac{1}{2} \cdot \ln\left(\frac{1 - \epsilon_3}{\epsilon_3}\right) = 1.018$$

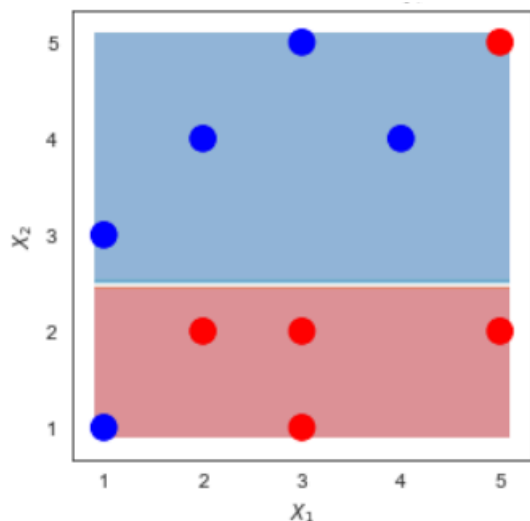
Row	x1	x2	y	Stump 3 Predicts	Weight 3
0	1	1	1	1	0.15385
1	1	3	1	1	0.03846
2	2	2	-1	1	0.03846 <-
3	2	4	1	1	0.16667
4	3	1	-1	1	0.03846 <-
5	3	2	-1	1	0.03846 <-
6	3	5	1	1	0.16667
7	4	4	1	1	0.16667
8	5	2	-1	-1	0.03846
9	5	5	-1	-1	0.15385

Summarise the results

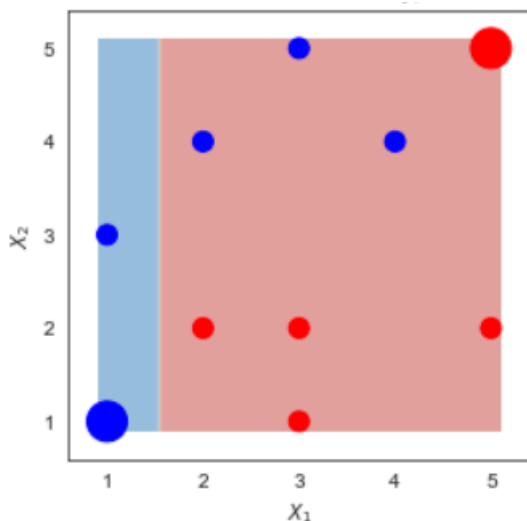
Row	x1	x2	y	Stump 1 Predicts	Weight 1	Weight 2	Stump 2 Predicts	Weight 3	Stump 3 Predicts
0	1	1	1	-1	0.1	0.25	1	0.153846	1
1	1	3	1	1	0.1	0.0625	1	0.0384615	1
2	2	2	-1	-1	0.1	0.0625	-1	0.0384615	1
3	2	4	1	1	0.1	0.0625	-1	0.166667	1
4	3	1	-1	-1	0.1	0.0625	-1	0.0384615	1
5	3	2	-1	-1	0.1	0.0625	-1	0.0384615	1
6	3	5	1	1	0.1	0.0625	-1	0.166667	1
7	4	4	1	1	0.1	0.0625	-1	0.166667	1
8	5	2	-1	-1	0.1	0.0625	-1	0.0384615	-1
9	5	5	-1	1	0.1	0.25	-1	0.153846	-1

Combining three Stumps

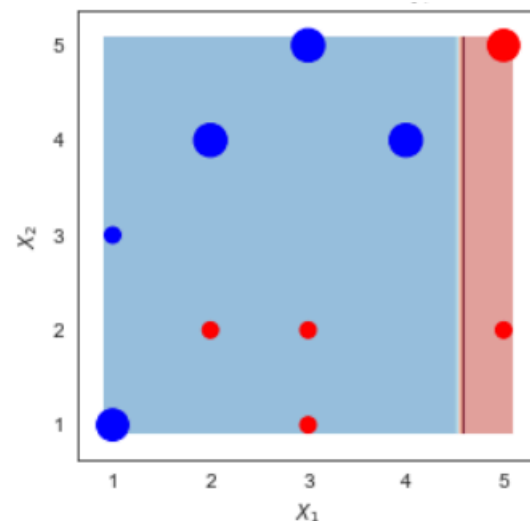
- Let say we stop making new stumps here.
- We will combine the three stumps to make the final model



$$\alpha_1 = .693$$



$$\alpha_2 = .733$$



$$\alpha_3 = 1.018$$

$$\text{Ada boost} = \text{sign} \left[\alpha_1 \cdot \text{Stump 1} + \alpha_2 \cdot \text{Stump 2} + \alpha_3 \cdot \text{Stump 3} \right]$$

$$H = \text{sign} \left[\alpha_1 \cdot \text{stump 1} + \alpha_2 \cdot \text{stump 2} + \alpha_3 \cdot \text{stump 3} \right]$$

$$= \text{sign} \left[.693 \cdot \underline{I(X_2 > 2.5)} + .733 \cdot \underline{I(X_1 < 1.5)} + 1.018 \cdot \underline{I(X_1 < 4.5)} \right]$$

(*) Region ① : $H = \text{sign} [.693 \cdot (-1) + .733 \cdot (1) + 1.018 \cdot (1)]$

$$= \text{sign} [1.058] = 1 \quad (\text{b/c } 1.058 > 0)$$

Region ② $H = \text{sign} [.693 * 1 + .733 * 1 + 1.018 * 1] = 1$

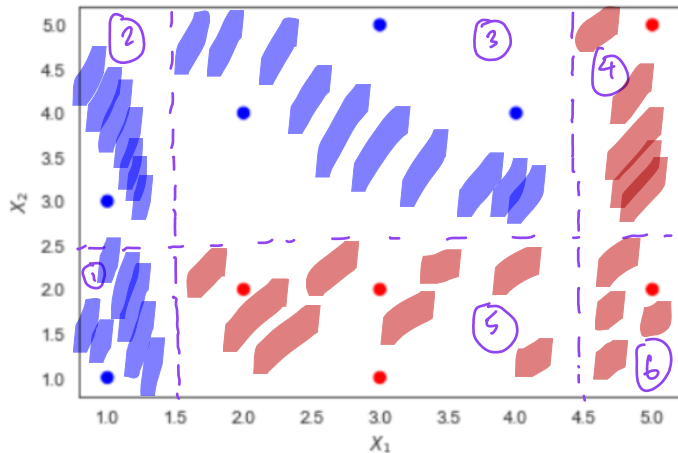
Region ③ :

$$H = \text{sign} [.93 * 1 + .733 * (-1) + 1.018 * 1]$$

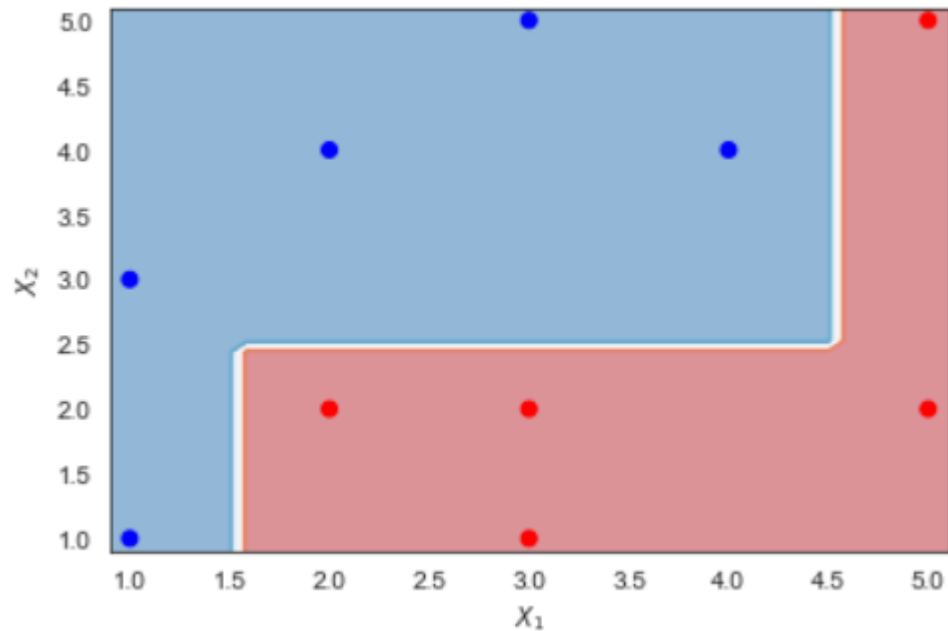
$$= \text{sign} [.98] = 1$$

Region ④ : $\text{sign} [.693 * 1 + .733(-1) + 1.018 * (-1)]$

$$= \text{sign} [-1.058] = -1$$



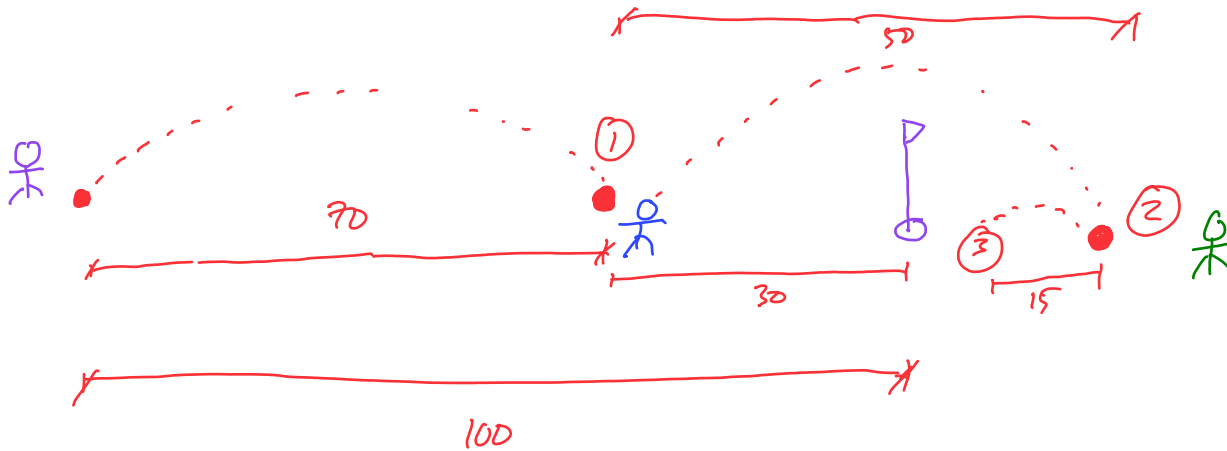
Combining three Stumps



The "key" idea of boosting method

- Train a **sequence** of **weak** models where the next model learns from the mistake / error of the current model.

- The final model is a combination of **All** the model in the sequence



Blue stick figure : Target : 100

reality : 70 $\Rightarrow$ error : $100 - 70 = 30$

Green stick figure : Target : 30

reality : 50 $\Rightarrow$ error : $30 - 50 = -20$

Target : target : -20 $\Rightarrow$ error : $-20 - (-15) = -5$

reality : -15

Data :

x_1 ↓ GPA	x_2 ↓ Age	y (target) ↓ Salary
3.5	60	100
4.0	29	120
3.0	49	80

train model 1 (weak model, says a stump



ϵ_1

GPA	Age	Salary	$\hat{y}_1$ (Model 1 predicts)	Model 1's error
3.5	60	100	90	$100 - 90 = 10$
4.0	29	120	150	$120 - 150 = -30$
3.0	49	80	70	$80 - 70 = 10$

train the second model, another stump



on the data

where the new target is ϵ_1

GPA	Age	target ϵ_1	Model 2 predicts $\hat{\epsilon}_1$	ϵ_2
3.5	60	10	5	5
4.0	29	-30	-20	-10
3.0	49	10	15	-5

we will fit the third model on the dataset where the target is ϵ_2 (error of the second model)

GPA	Age	ϵ_2 target	Model 3 predicts $\hat{\epsilon}_2$
3.5	60	5	5
4.0	29	-10	-8
3.0	49	-5	-6

⇒ we combine the final predictions of the 3 models

as:

$$H = \text{Predictions of Model 1} + \text{Predictions of M2} + \text{Pred. of M3}$$

$$= \hat{y}_1 + \hat{\epsilon}_1 + \hat{\epsilon}_2$$

GPA	Age	Salary Y	$\hat{y}_1$ (Model 1 predicts)	$\hat{\epsilon}_1$	$\hat{\epsilon}_2$	Final pred
3.5	60	100	90	5	5	100
4.0	29	120	150	-20	-8	122
3.0	49	80	70	15	-6	79

Learning rate